



Assignment: Using Packages, Inheritance, and Polymorphism

Objective

To understand and implement packages in Java, along with inheritance across packages, method overriding, and polymorphism, using proper object-oriented design practices.

Problem Statement

You are required to develop a Java application using three different packages.

The application must demonstrate:

- Class creation inside packages
- Inheritance across packages
- Runtime polymorphism using superclass references
- Proper use of constructors, getters/setters, and overridden methods

Package Structure (Mandatory)

You must create the following packages exactly as specified:

- p1
- p2
- p3

Package p1

Class: Employee

This class acts as the base (super) class.

Attributes

- id
- name
- salary

Methods

- Parameterized constructor
- Getter and setter methods for all attributes
- toString() method
- calSal() method (abstract method)

Requirements

- Employee must act as the base class
- The calSal() method must be overridable by subclasses
- Use appropriate access modifiers so that subclasses in another package can access required members

Package p2

Create the following classes in this package:

- HR
- SalesManager
- Admin

All classes in this package must extend Employee.

Class-Specific Requirements

HR

- Additional attribute:
 - commission



SalesManager

- Additional attributes:
 - incentive
 - target

Admin

- Additional attribute:
 - allowance

Common Requirements for All Classes in p2

Each class must define:

- Parameterized constructor
- Getter and setter methods
- Overridden `toString()` method
- Overridden `calSal()` method with role-specific salary calculation logic

Package p3

Class: Test

This class represents the application entry point.

Requirements

- Must contain the `main()` method
- Import required classes from packages p1 and p2
- Create an array of type `Employee`
- Store objects of:
 - HR
 - SalesManager
 - Admin
- Iterate through the array and:
 - Call the polymorphic method `calSal()`
 - Display object details using `toString()`

Task Requirements

1. Use the package keyword correctly in all classes
2. Use import statements wherever required
3. Demonstrate inheritance across packages
4. Demonstrate runtime polymorphism using an `Employee[]`
5. Use proper access modifiers (`public`, `protected`, `default`) appropriately
6. Follow standard Java coding conventions

Important Rules

- Package names must be exactly p1, p2, and p3
- Do not place any class in the default package
- Every class must contain:
 - Attributes
 - Constructor
 - Getters and setters
 - `toString()`
 - `calSal()`
- Code must compile and run successfully